

Course 5022

Semester V

Theory

4 Credits

# Design of Machine Elements

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

|                 |                                      |
|-----------------|--------------------------------------|
| Programmes      | <b>Mechanical Engineering</b>        |
| Contact hours   | <b>60</b>                            |
| Teaching scheme | <b>3:1:0:0</b>                       |
| Credits         | <b>4</b>                             |
| CIE / ESE       | <b>Theory CIE 40 / Theory ESE 60</b> |
| Self-learning   | <b>5.5 h/week</b>                    |

## CURRICULUM INTEGRITY

# Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5022 as Design of Machine Elements in Semester V for Mechanical Engineering. The official SITTTR detailed source was unavailable. With user approval, this handbook uses MIT OpenCourseWare's Elements of Mechanical Design syllabus as a reliable supplementary source and does not represent recovered official REV2026 detailed-syllabus wording.

**Source treatment:** Teaching scheme, credits and assessment values are provisional placeholders. This handbook cannot be used to fabricate or operate a component without current standards, verified loads/material data, tolerance/drawing review and qualified engineering approval.

**Verified source note:** The source supports bearings, springs, gears, mechanisms, modeling, design integration, fabrication/testing and responsible design practice.

## COURSE DASHBOARD

# What You Are Expected to Learn

**60**

Contact hours

**4**

Credits

**Theory CIE 40**

CIE

**Theory ESE 60**

ESE

## Course introduction

Design of Machine Elements applies mechanics, materials, standards and design judgment to components such as joints, shafts, keys, couplings, bearings, springs and gears.

**Malayalam support:** ເປັນ handbook ທີ່ອະນຸຍາດໃຫ້ສູນກາງສຶກສາ ສະຫຼຸບ ກົດເກນ ກົດເກນ ກົດເກນ; ກົດເກນ ກົດເກນ principle, diagram/procedure, application, common mistake ກົດເກນ ກົດເກນ ກົດເກນ.

## Prerequisite foundation

Strength of materials, engineering mechanics, machine drawing and current design-code/data access.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

### Teaching and assessment scheme

| L | T | P | S | Self-learning/week | Contact hours | Credits | CIE           | ESE           |
|---|---|---|---|--------------------|---------------|---------|---------------|---------------|
| 3 | 1 | 0 | 0 | 5.5 h/week         | 60            | 4       | Theory CIE 40 | Theory ESE 60 |

### STUDY METHOD

## How to Use This Handbook

### 1 • Understand

Read terms, conditions and purpose; make a short definition in your own words.

### 2 • Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

### 3 • Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

## 4 • Revise

Use questions, quick cards and common-mistake notes before examination.

### OFFICIAL STATEMENTS

## Objectives and Course Outcomes

### Official course objectives

1. Explain design process, loads, stress and failure considerations.
2. Apply conceptual design sequences to joints, shafts and couplings.
3. Explain selection/design principles for bearings, springs and gears.
4. Communicate assumptions, checks, manufacturability and safety limits of a design.

### Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

**Malayalam support:** CO ക്കായിട്ടുള്ള exam-ൽ ഉപയോഗിക്കേണ്ട പ്രധാനപ്പെട്ട പദങ്ങൾ ഉപയോഗിച്ച്. Define, explain, apply എന്നിവയ്ക്കുള്ള action words ഉപയോഗിക്കേണ്ടതാണ്.

### Official Course Outcomes

| CO  | Official outcome                                                                | Bloom level |
|-----|---------------------------------------------------------------------------------|-------------|
| CO1 | Explain machine-design process, loading, material and failure considerations.   | Understand  |
| CO2 | Apply approved design-sequence concepts for joints, shafts, keys and couplings. | Apply       |
| CO3 | Explain gear, bearing and spring selection/design principles.                   | Understand  |
| CO4 | Document design assumptions, checks and limitations responsibly.                | Apply       |

### Official CO-PO articulation matrix

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7-PO11 |
|-----|-----|-----|-----|-----|-----|-----|----------|
| CO1 | 3   | 2   | 2   | 2   | 2   | -   | -        |
| CO2 | 3   | 2   | 2   | 2   | 2   | -   | -        |
| CO3 | 3   | 2   | 2   | 2   | 2   | -   | -        |
| CO4 | 3   | 2   | 2   | 2   | 2   | -   | -        |

### SYLLABUS COVERAGE

## Curriculum Coverage Map

### All official major topics mapped to handbook chapters

| Module | Title                                   | Official major topics                                                                                               | Hours            | CO  | Handbook section |
|--------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------|-----|------------------|
| 1      | Design Process, Loads and Materials     | Design requirements, load paths, stress, fatigue, material properties, factor of safety and manufacturability.      | Approx. 14 hours | CO1 | Module 1         |
| 2      | Joints, Shafts and Couplings            | Fasteners/welds, shafts/axles, keys, splines and couplings; loading, stress concentration and safe design sequence. | Approx. 16 hours | CO2 | Module 2         |
| 3      | Gears, Bearings and Springs             | Power transmission, gear geometry, rolling/plain bearings, lubrication, spring behaviour and selection.             | Approx. 16 hours | CO3 | Module 3         |
| 4      | Integration, Testing and Responsibility | Assemblies, tolerance stack-up, prototype/test, failure investigation, documentation and safety.                    | Approx. 14 hours | CO4 | Module 4         |

### LEARNING ROADMAP

## How the Modules Connect

### Design Process, Loads and Materials

Design requirements, load paths, stress, fatigue, material properties, factor of safety and manufacturability.

CO1 · Approx. 14 hours

## Joists, Shafts and Couplings

Fasteners/welds, shafts/axles, keys, splines and couplings; loading, stress concentration and safe design sequence.

CO2 · Approx. 16 hours

## Gears, Bearings and Springs

Power transmission, gear geometry, rolling/plain bearings, lubrication, spring behaviour and selection.

CO3 · Approx. 16 hours

## Integration, Testing and Responsibility

Assemblies, tolerance stack-up, prototype/test, failure investigation, documentation and safety.

CO4 · Approx. 14 hours

### MODULE 1

## Design Process, Loads and Materials

Design requirements, load paths, stress, fatigue, material properties, factor of safety and manufacturability.

Approx. 14 hours

CO1

Understand · Apply

### Learning goals

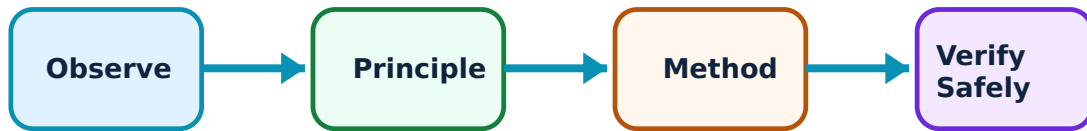
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Design Process, Loads and Materials: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. Engineering design process–

A design begins with function, constraints, loads, environment and interfaces, then proceeds through modelling, selection, analysis, prototyping/testing and review.

#### Study and application method

State function, load cases, material/environment, life, safety factor basis, constraints and validation plan before calculation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

#### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** State function, load cases, material/environment, life, safety factor basis, constraints and validation plan before calculation.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept ഈ problem ഈ ഏതു ഏതു. ഈ diagram / circuit / formula / procedure ഈ ഏതു. ഈ result ഈ application ഈ ഏതു. Technical terms English-ഈ ഈ.

**Common mistake / precaution:** A spreadsheet or CAD model is not validation without correct inputs and independent review.

## 2. Stress and failure awareness–

Static, fluctuating, impact and thermal loads create normal/shear stress and potential yielding, fracture, fatigue, wear or buckling.

### Study and application method

Draw free-body/load path, identify critical section and relevant failure mode; use the prescribed theory/data with units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Draw free-body/load path, identify critical section and relevant failure mode; use the prescribed theory/data with units.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept ഈ പ്രശ്നം പരിഹരിക്കാൻ ഉപയോഗിക്കണം. ഈ diagram / circuit / formula / procedure ഉപയോഗിക്കണം. ഈ result ഉപയോഗിക്കണം application ഉപയോഗിക്കണം. Technical terms English-ൽ ഉപയോഗിക്കണം.

**Common mistake / precaution:** Do not combine incompatible material properties, units or safety factors.

## 3. Material and manufacturing choice–

Selection balances strength, stiffness, toughness, corrosion, cost, availability, process and inspection. Tolerances and surface finish affect fit and fatigue.

### Study and application method

Compare candidates against stated criteria and record source/version of material data.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Compare candidates against stated criteria and record source/version of material data.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Never substitute material or heat treatment without approved engineering change control.

**Module 1 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

## MODULE 2

# Joints, Shafts and Couplings

Fasteners/welds, shafts/axles, keys, splines and couplings; loading, stress concentration and safe design sequence.

Approx. 16 hours

CO2

Understand · Apply

## Learning goals

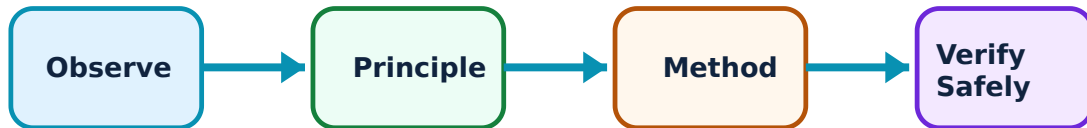
Identify core terms, explain the principle and complete the stated application or practical outcome.

## Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Joints, Shafts and Couplings: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Joints and fasteners–

Bolted, welded and keyed joints transmit forces while allowing assembly/service. Preload, load distribution, fatigue, corrosion and inspection influence reliability.

### Study and application method

Identify joint load path, failure modes and required design/inspection checks under the approved standard.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Identify joint load path, failure modes and required design/inspection checks under the approved standard.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept എങ്ങനെ എങ്ങനെ എങ്ങനെ. എന്ത് diagram / circuit / formula / procedure എങ്ങനെ എങ്ങനെ. എങ്ങനെ result എങ്ങനെ application എങ്ങനെ എങ്ങനെ എങ്ങനെ. Technical terms English-  
എങ്ങനെ എങ്ങനെ.

**Common mistake / precaution:** Do not use generic bolt torque or weld size without drawing, grade and procedure data.

## 2. Shafts, keys and couplings–

Shafts transmit torque often with bending; keys/couplings transfer torque and must avoid fretting, fatigue and misalignment problems.

### Study and application method

Determine torque/bending/load cases, critical geometry and prescribed equivalent-stress/twist checks; document keyway stress-concentration consideration.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Determine torque/bending/load cases, critical geometry and prescribed equivalent-stress/twist checks; document keyway stress-concentration consideration.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** കോൺസെപ്റ്റ് വ്യക്തമായി വിശദീകരിക്കുക. ഡയഗ്രാം / സർക്യൂട്ട് / ഫോർമുല / പ്രോസീഡർ ഉപയോഗിക്കുക. ഫലം വ്യക്തമായി വിശദീകരിക്കുക. Technical terms English-ൽ ഉപയോഗിക്കുക.

**Common mistake / precaution:** Rotating machinery requires guarding, balance and qualified assembly; design exercise values are not operating approval.

## 3. Design verification–

Design verification cross-checks load, stress, deflection, critical speed, fits, assembly and test plan against requirements.

### Study and application method

Create a checklist of assumptions, governing case, units, code/data source, result and independent sanity check.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Create a checklist of assumptions, governing case, units, code/data source, result and independent sanity check.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept ഈ പ്രശ്നം പരിഹരിക്കാൻ ഉപയോഗിക്കണം. ഈ diagram / circuit / formula / procedure ഈ പ്രശ്നം പരിഹരിക്കാൻ ഉപയോഗിക്കണം. ഈ result ഈ പ്രശ്നം പരിഹരിക്കാൻ ഉപയോഗിക്കണം. Technical terms English-ൽ ഉപയോഗിക്കണം.

**Common mistake / precaution:** Do not release a component before peer/qualified review.

**Module 2 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

### MODULE 3

## Gears, Bearings and Springs

Power transmission, gear geometry, rolling/plain bearings, lubrication, spring behaviour and selection.

Approx. 16 hours

CO3

Understand · Apply

### Learning goals

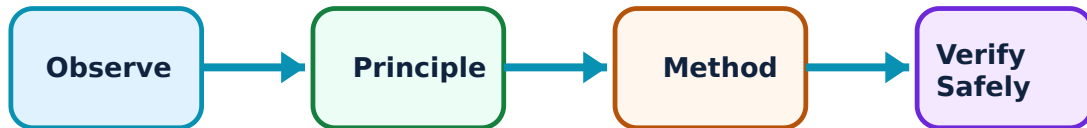
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Gears, Bearings and Springs: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Gears and drives–

Gears transmit motion/torque with defined ratio; tooth strength, wear, alignment, lubrication and housing stiffness affect life/noise.

### Study and application method

State ratio, power/speed, duty, load factor, material and required standard method; sketch force directions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** State ratio, power/speed, duty, load factor, material and required standard method; sketch force directions.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept നോക്കണം എന്ന് തിരിച്ചറിയണം. ഏതു diagram / circuit / formula / procedure ഉപയോഗിക്കണം. ഏതു result നേടണം application നോക്കണം. Technical terms English-ൽ എഴുതണം.

**Common mistake / precaution:** Do not place hands near rotating drives; use guards and lockout before inspection.

## 2. Bearings and lubrication–

Bearings support radial/axial load while reducing friction. Selection depends on load, speed, life, alignment, lubrication, contamination and mounting.

### Study and application method

Identify bearing load/direction, operating environment and life requirement; follow catalogue/standard selection procedure.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Identify bearing load/direction, operating environment and life requirement; follow catalogue/standard selection procedure.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept എങ്ങനെ എങ്ങനെ എങ്ങനെ. diagram / circuit / formula / procedure എങ്ങനെ. result application എങ്ങനെ. Technical terms English-എങ്ങനെ.

**Common mistake / precaution:** Wrong lubrication, preload or fit can rapidly damage a bearing.

## 3. Springs–

Springs store energy and provide force/deflection. Stress, fatigue, buckling, solid height and end condition are key considerations.

### Study and application method

State load range, deflection, material, end condition and required checks under prescribed method.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** State load range, deflection, material, end condition and required checks under prescribed method.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Compressed springs store energy; use controlled handling and guarding.

**Module 3 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

## MODULE 4

# Integration, Testing and Responsibility

Assemblies, tolerance stack-up, prototype/test, failure investigation, documentation and safety.

Approx. 14 hours

CO4

Understand · Apply

## Learning goals

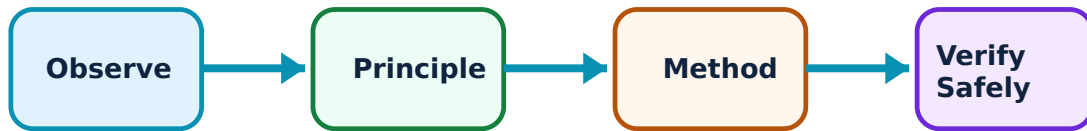
Identify core terms, explain the principle and complete the stated application or practical outcome.

## Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Integration, Testing and Responsibility: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. System integration–

Machine elements interact through interfaces, fits, stiffness, thermal expansion, lubrication and assembly sequence.

#### Study and application method

Map interfaces and tolerance/assembly risks; plan inspection and functional test.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

#### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Map interfaces and tolerance/assembly risks; plan inspection and functional test.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept ന്നുള്ളതാണ് ചോദ്യം. ഏതു diagram / circuit / formula / procedure ഉപയോഗിക്കണം. ഏതു result ന്നുള്ളതാണ് application. Technical terms English-ൽ എഴുതണം.

**Common mistake / precaution:** Local component adequacy does not guarantee system safety.

## 2. Testing and failure learning–

Tests validate assumptions under controlled conditions; failure analysis identifies mechanism and corrective action.

### Study and application method

Define objective, instrumentation, acceptance criterion, controls and stop conditions before testing.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Define objective, instrumentation, acceptance criterion, controls and stop conditions before testing.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

**Malayalam support:** ഏതു concept ന്നുള്ള ഉത്തരം നൽകേണ്ടതാണ്. ഏതു diagram / circuit / formula / procedure ന്നുള്ള ഉത്തരം നൽകേണ്ടതാണ്. ഏതു result ന്നുള്ള ഉത്തരം നൽകേണ്ടതാണ്. Technical terms English-ൽ ഉത്തരം നൽകേണ്ടതാണ്.

**Common mistake / precaution:** Never test beyond safe enclosure/load limits.

## 3. Design documentation–

Drawings, calculations, BOM, specifications, revisions and test records preserve design intent and traceability.

### Study and application method

Issue a complete controlled package with assumptions, units, revision and approval status.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## How to construct a complete answer

**Given:** identify the quantities, device condition, waveform or circuit arrangement.

**Required:** state exactly what must be found or explained.

**Method:** Issue a complete controlled package with assumptions, units, revision and approval status.

**Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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**Common mistake / precaution:** Do not fabricate from uncontrolled or ambiguous drawings.

**Module 4 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

## FORMULA AND PROCEDURE BANK

### Rapid Recall Bank

#### Recall 1

State symbols and SI units before substitution.

#### Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

### Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

### Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

### Recall 5

For a comparison: use construction, working, result, advantage and limitation.

### Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

#### DIAGRAM LIBRARY

### Diagram Library for Rapid Revision

#### Module 1: Design Process, Loads and Materials

Use the concept flow to revise observation, principle, method and verification.

#### Module 2: Joints, Shafts and Couplings

Use the concept flow to revise observation, principle, method and verification.

#### Module 3: Gears, Bearings and Springs

Use the concept flow to revise observation, principle, method and verification.

## Module 4: Integration, Testing and Responsibility

Use the concept flow to revise observation, principle, method and verification.

### SELF-LEARNING

## Official Self-Learning and Student Activities

### Structured activity

Prepare a load-path and failure-mode table for a shaft-coupling assembly.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Compare two bearing options using stated catalogue criteria.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Create a design-review checklist for a spring or gear exercise.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Document a conceptual prototype test plan with safety controls.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

**Activity discipline:** The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

# Assessment Methodology and Examination Pattern

| Official assessment structure |                            |                                                       |
|-------------------------------|----------------------------|-------------------------------------------------------|
| Component                     | Official mark / conversion | Student evidence                                      |
| Attendance                    | 5 marks                    | End-of-semester attendance component.                 |
| Self-learning activities      | 15 marks                   | Module-linked activities.                             |
| Series examinations           | 20 + 20 converted          | Two 50-mark series examinations.                      |
| CIE total                     | Theory CIE 40              | Continuous internal evaluation.                       |
| Written ESE                   | Theory ESE 60              | 2.5 hours; official Part A / Part B / Part C pattern. |

## Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

## Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

## EXAM PRACTICE

## Module-wise Questions with Answers

### Module 1 — Design Process, Loads and Materials

**Explain Engineering design process and state one correct method, application or precaution.—**

A design begins with function, constraints, loads, environment and interfaces, then proceeds through modelling, selection, analysis, prototyping/testing and review.

**Answer framework:** State function, load cases, material/environment, life, safety factor basis, constraints and validation plan before calculation.

**Important point:** A spreadsheet or CAD model is not validation without correct inputs and independent review.

**Explain Stress and failure awareness and state one correct method, application or precaution. –**

Static, fluctuating, impact and thermal loads create normal/shear stress and potential yielding, fracture, fatigue, wear or buckling.

**Answer framework:** Draw free-body/load path, identify critical section and relevant failure mode; use the prescribed theory/data with units.

**Important point:** Do not combine incompatible material properties, units or safety factors.

**Explain Material and manufacturing choice and state one correct method, application or precaution. –**

Selection balances strength, stiffness, toughness, corrosion, cost, availability, process and inspection. Tolerances and surface finish affect fit and fatigue.

**Answer framework:** Compare candidates against stated criteria and record source/version of material data.

**Important point:** Never substitute material or heat treatment without approved engineering change control.

## Module 2 — Joints, Shafts and Couplings

**Explain Joints and fasteners and state one correct method, application or precaution. –**

Bolted, welded and keyed joints transmit forces while allowing assembly/service. Preload, load distribution, fatigue, corrosion and inspection influence reliability.

**Answer framework:** Identify joint load path, failure modes and required design/inspection checks under the approved standard.

**Important point:** Do not use generic bolt torque or weld size without drawing, grade and procedure data.

**Explain Shafts, keys and couplings and state one correct method, application**

### or precaution.–

Shafts transmit torque often with bending; keys/couplings transfer torque and must avoid fretting, fatigue and misalignment problems.

**Answer framework:** Determine torque/bending/load cases, critical geometry and prescribed equivalent-stress/twist checks; document keyway stress-concentration consideration.

**Important point:** Rotating machinery requires guarding, balance and qualified assembly; design exercise values are not operating approval.

### Explain Design verification and state one correct method, application or precaution.–

Design verification cross-checks load, stress, deflection, critical speed, fits, assembly and test plan against requirements.

**Answer framework:** Create a checklist of assumptions, governing case, units, code/data source, result and independent sanity check.

**Important point:** Do not release a component before peer/qualified review.

## Module 3 — Gears, Bearings and Springs

### Explain Gears and drives and state one correct method, application or precaution.–

Gears transmit motion/torque with defined ratio; tooth strength, wear, alignment, lubrication and housing stiffness affect life/noise.

**Answer framework:** State ratio, power/speed, duty, load factor, material and required standard method; sketch force directions.

**Important point:** Do not place hands near rotating drives; use guards and lockout before inspection.

### Explain Bearings and lubrication and state one correct method, application or precaution.–

Bearings support radial/axial load while reducing friction. Selection depends on load, speed, life, alignment, lubrication, contamination and mounting.

**Answer framework:** Identify bearing load/direction, operating environment and life requirement; follow catalogue/standard selection procedure.

**Important point:** Wrong lubrication, preload or fit can rapidly damage a bearing.

### **Explain Springs and state one correct method, application or precaution.–**

Springs store energy and provide force/deflection. Stress, fatigue, buckling, solid height and end condition are key considerations.

**Answer framework:** State load range, deflection, material, end condition and required checks under prescribed method.

**Important point:** Compressed springs store energy; use controlled handling and guarding.

## **Module 4 — Integration, Testing and Responsibility**

### **Explain System integration and state one correct method, application or precaution.–**

Machine elements interact through interfaces, fits, stiffness, thermal expansion, lubrication and assembly sequence.

**Answer framework:** Map interfaces and tolerance/assembly risks; plan inspection and functional test.

**Important point:** Local component adequacy does not guarantee system safety.

### **Explain Testing and failure learning and state one correct method, application or precaution.–**

Tests validate assumptions under controlled conditions; failure analysis identifies mechanism and corrective action.

**Answer framework:** Define objective, instrumentation, acceptance criterion, controls and stop conditions before testing.

**Important point:** Never test beyond safe enclosure/load limits.

### **Explain Design documentation and state one correct method, application or precaution.–**

Drawings, calculations, BOM, specifications, revisions and test records preserve design intent and traceability.

**Answer framework:** Issue a complete controlled package with assumptions, units, revision and approval status.

**Important point:** Do not fabricate from uncontrolled or ambiguous drawings.

#### PRACTICE ONLY

## Practice Model Paper

**Important:** This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTR model question paper.

### Part A — short response

1. State one key definition from Module 1: Design Process, Loads and Materials.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Joints, Shafts and Couplings.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Gears, Bearings and Springs.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Integration, Testing and Responsibility.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

### Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

### Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Design requirements, load paths, stress, fatigue, material properties, factor of safety and manufacturability..
2. Develop a complete answer or practical plan for: Fasteners/welds, shafts/axles, keys, splines and couplings; loading, stress concentration and safe design

sequence..

3. Develop a complete answer or practical plan for: Power transmission, gear geometry, rolling/plain bearings, lubrication, spring behaviour and selection..
4. Develop a complete answer or practical plan for: Assemblies, tolerance stack-up, prototype/test, failure investigation, documentation and safety..

#### LAST-MILE REVISION

## Quick Revision

### Module 1: Design Process, Loads and Materials

Design requirements, load paths, stress, fatigue, material properties, factor of safety and manufacturability.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 2: Joints, Shafts and Couplings

Fasteners/welds, shafts/axles, keys, splines and couplings; loading, stress concentration and safe design sequence.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 3: Gears, Bearings and Springs

Power transmission, gear geometry, rolling/plain bearings, lubrication, spring behaviour and selection.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 4: Integration, Testing and Responsibility

Assemblies, tolerance stack-up, prototype/test, failure investigation, documentation and safety.

- Match each topic to CO4.

- Redraw or rehearse one key method.
- Check one common mistake/precaution.

**Common mistakes:** Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

**Exam checklist:** Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

## VOCABULARY

# Glossary

## Important terms from the syllabus

| Term / topic                      | One-line meaning                                                                                                                                                     |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Engineering design process        | A design begins with function, constraints, loads, environment and interfaces, then proceeds through modelling, selection, analysis, prototyping/testing and review. |
| Stress and failure awareness      | Static, fluctuating, impact and thermal loads create normal/shear stress and potential yielding, fracture, fatigue, wear or buckling.                                |
| Material and manufacturing choice | Selection balances strength, stiffness, toughness, corrosion, cost, availability, process and inspection.                                                            |
| Joints and fasteners              | Bolted, welded and keyed joints transmit forces while allowing assembly/service.                                                                                     |
| Shafts, keys and couplings        | Shafts transmit torque often with bending; keys/couplings transfer torque and must avoid fretting, fatigue and misalignment problems.                                |
| Design verification               | Design verification cross-checks load, stress, deflection, critical speed, fits, assembly and test plan against requirements.                                        |
| Gears and drives                  | Gears transmit motion/torque with defined ratio; tooth strength, wear, alignment, lubrication and housing stiffness affect life/noise.                               |
| Bearings and lubrication          | Bearings support radial/axial load while reducing friction.                                                                                                          |
| Springs                           | Springs store energy and provide force/deflection.                                                                                                                   |
| System integration                | Machine elements interact through interfaces, fits, stiffness, thermal expansion, lubrication and assembly sequence.                                                 |
| Testing and failure learning      | Tests validate assumptions under controlled conditions; failure analysis identifies mechanism and corrective action.                                                 |
| Design documentation              | Drawings, calculations, BOM, specifications, revisions and test records preserve design intent and traceability.                                                     |

## LEARNING RESOURCES

### References

#### Recommended machine-element-design references

1. Joseph Shigley et al., Mechanical Engineering Design.
2. R. L. Norton, Design of Machinery.
3. Machinery's Handbook.

#### Approved public machine-design source

- <https://ocw.mit.edu/courses/2-72-elements-of-mechanical-design-spring-2009/pages/syllabus/>

This source supports the study handbook while official Course 5022 detail is unavailable; it does not replace standards or qualified design review.